

Clinical Practice Guideline (CPG) for Autism Spectrum Disorder (ASD)

Table 1. Description of the CPG-Targeted Population and Its Intended Users

Attribute	Description
Disease/Condition	Autism Spectrum Disorder (ASD)
Target Population	Children under the age of 16
Clinical Specialty (Intended Users)	Pediatrics, Family Medicine Physicians, Child and Adolescent Psychiatry, Neurodevelopmental Pediatrics, Pediatric Neurology, Clinical Psychology, Occupational Therapy, Speech and Language Pathology, Special Education, Nursing.

Recommendations:

- **Multidisciplinary Function:** A diagnostic and specialty assessment, which is carried out by an experienced multidisciplinary team.
- **The Most Responsible Physician (MRP):** The MRP reviews the care plans of different disciplines working with children on the autism spectrum, ensuring that those plans are integrated, continuous, and coordinated.
- **Qualifications:**
 - A preliminary clinical diagnosis using the DSM-5 criteria can be made by child psychologists, psychiatrists, and general pediatricians, who are suitably trained and skilled in the application of those criteria.
 - The diagnostic evaluation team should include a neurodevelopmental or developmental-behavioral pediatrician, child psychiatrist, or pediatric neurologist.
 - A consultant clinical psychologist (Ph.D. or PsyD. holder) acts as a core member of the diagnostic team: A psychologist with doctoral-level training in psychology, which includes psychological testing supervision.
 - Services are overseen by clinical psychologists (Master's degree level) with experience and expertise in ASD and can seek supervision from consultant clinical psychologists directly, online, or by phone.
 - The behavioral intervention process must be overseen by a professional with a minimum qualification of an M.Sc. in clinical psychology or a related field; it is preferable that she/he holds either a Board Certified Behavior Analyst (BCBA, BCBA-D) or a Behavior Analyst Certification Board (BACB) certificate.
 - The interventionist requires a B.Sc. in psychology or a similar field with courses related to child psychology, disability, child development, and applied behavior analysis (ABA).
 - Occupational therapy (OT) should be provided by a practitioner with an OT B.Sc. or M.Sc., at least five years of clinical experience in pediatric settings, and a license from the Saudi Commission for Health Specialties (SCFHS).
 - Speech and language pathologists (SLPs) with the requisite expertise can contribute to the diagnosis of ASD, normally through joint-working with professionals from other disciplines in an integrated team.

Table 2. Recommendations for Assessment

Medical Assessment

- A medical assessment is recommended for every child on the autism spectrum. Normally, that assessment consists of physical tests (including hearing), screening for dysmorphic features, and a neurocutaneous examination to identify any indicators of tuberous sclerosis.
- It is also essential to combine observing the child's current symptoms with a detailed study of his/her record of behavior.
- That record needs to take into account how the child's symptoms have developed over time, including assessing how they have impacted his/her functioning within his/her family, among peers, and/or within school or care center settings.
- The diagnostic process must include formal assessments of the child's sensory status and his/her adaptive, cognitive, and language capabilities.
- Both the Social Communication Questionnaire (SCQ) and the Modified Checklist for Autism in Toddlers (M-CHAT) can be used as screening tools for the assessment of ASD.
- Various questionnaires can be used to review the child's history of ASD symptoms, including the Social Communication Questionnaire (SCQ), the Social Responsiveness Scale (SRS), and the revised version of the Autism Diagnostic Interview (ADI-R).
- Similarly, a diagnosis can be confirmed by collecting data through the use of validated tools, such as the second editions of the Childhood Autism Rating Scale (CARS-2) and the Autism Diagnostic Observation Schedule (ADOS-2).
- The tests referred to above should also incorporate the analysis of the child's physical development (such as the circumference of their head) compared to standard trajectories, identification of any abnormal differences in body structure or organ size, checking for external evidence of neurocutaneous syndromes, and assessing for neurologic disorders.
- Assessment of children who have been diagnosed with ASD should include screening for conditions that frequently manifest alongside or have an etiological relationship with ASD.
- As part of the screening for etiological conditions, all families should be advised to have genetic testing and be provided with the opportunity to have such testing.
- In cases in which the contributing (i.e., etiological) conditions for a specific developmental issue are unknown, a chromosome microarray should be undertaken.
- The overall assessment should include evaluating whether the individual has any of the various neurological conditions that commonly afflict children on the autism spectrum, including disorders related to behavior, anxiety, feeding, self-harm, parental attachment, tics, and ADHD. If the assessment reveals the presence of any such disorders, it should be followed by a comprehensive evaluation of those conditions.
- When it has been identified that the child may have a particular disorder or syndrome, the professional making the assessment should either apply the relevant tests or refer the child to the appropriate specialist.

Psychological Assessment

- Selecting appropriate methods for conducting this assessment, including but not limited to tests, interviews, observations, surveys, or other data-gathering techniques.
- Assessing strong points, capabilities, and protective factors that lower the risk of negative outcomes.
- Establishing a comprehensive developmental and family history.
- Completing a thorough cognitive assessment, potentially including psychometrics if required.
- Assessing learning styles, strong points, and any issues impeding learning.
- Assessing families' needs.

Behavioral Assessment

- Determining the current level of development by using criterion-referenced and normed-referenced tools (e.g., the Adaptive Behavior Assessment System or the Vineland adaptive behavior scale).
- Developing an individualized program that focuses on all developmental domains to address current needs.

Occupational Therapy Assessment

- With reference to the DSM-5, assess any issues with the processing of sensory information, including hyper/hyposensitivity.
- Cognitive and intellectual functioning indicating atypical development.
- Identify any functional/adaptive challenges that restrict the child's ability to live fully (e.g., issues relating to sleep, education, personal care, play, safety, or community engagement).
- Motor development.
- Academic skills.
- Food-related problems, including restricted food intake, dietary problems, or mealtime difficulties.
- Sexuality.
- Analyze occupational performance to highlight any positive or negative issues related to participation and performance (including evaluating the child's capabilities and typical patterns of performance alongside the impact of different environmental settings, specific activities, and client factors. All the above should be measured objectively through both non-standardized and standardized assessments).

Speech–Language Assessment

- Thoroughly assessing speech and language also includes examining capabilities related to swallowing and eating generally.
- If required, an assessment of augmentative and alternative communication (AAC) should be undertaken to identify possible benefits related to enhancing the child's communicative capacities.
- Identifying participative and activity-related limitations, taking account of the broader health implications of such (e.g., hydrating and having adequate nutrition) and their consequences for the child's participation in such standard events as meals and family celebrations.
- Highlighting any issues relating to the individual or their environment that either impede or enhance their consumption of adequate nutrition (e.g., their dietary preferences and familial inputs to implement approaches to safe drinking and eating).
- Judging the extent to which problems related to feeding and swallowing impact the child's life and the lives of their family.
- Such evaluations should take account of the child's needs, including such issues as facilitating communication through identifying alternative means, augmenting natural speech, or developing communicative practices that encourage more suitable behavior.

Table 3. Recommendations for Intervention

Medical Interventions

- All interventions should be personalized, appropriate to the child's level of development, suitably intensive, and measured using data that enable the impact of the intervention to be evaluated and its course to be adjusted in line with its overall objective.
- The prescribing clinician must thoroughly consider both the advantages and disadvantages of using pharmaceuticals to address behavioral issues and should only use such interventions as part of a more holistic approach.

- The consideration of the advantages and disadvantages of prescription medicine must be based on an understanding of each drug's required doses, indications and contraindications, potential side effects, relevant drug–drug interactions, and the monitoring required for its usage.
- Prescription medication can be used to manage a range of conditions and symptoms, including irritability and challenging behaviors.
- All therapeutic interventions should be informed by the correct diagnosis of any psychiatric issues that manifest alongside ASD.
- The medicinal intervention pathway should only be taken following the identification of when the behavior commenced and a thorough assessment of factors that trigger it or make it worse.

Psychological Interventions

- Involve the parents/caregivers, starting with providing them the chance to witness the evaluation process and talk about the detail of their child's behaviors with the practitioners following that evaluation.
- Discuss with the parents their child's strong points, areas for development, and the intervention program that will be delivered so that they can be appropriately engaged in coordinating and advocating for that program.

Behavioral Intervention

- The ABA principles inform the majority of the intervention models that are based upon evidence of what works.
- Aspects of the ABA are combined with other principles derived from naturalistic developmental behavioral interventions (NDBIs). Such other NDBI principles include working towards targets for learning and skill acquisition that are developmentally appropriate and delivering interventions within social and environmental settings that are natural.
- Plans are based on a rigorous comprehensive assessment.
- Plans should focus on teaching new skills and reducing maladaptive behaviors.
- Plans should consist of goals that meet the criteria of being specific, measurable, achievable, relevant, and time-bound (i.e., SMART goals).
- Intensive early intervention programs should be used for children between 2 and 5 years old.
- Focused interventions and/or parent-mediated interventions may address severe problems and behaviors.
- Focused interventions might be the model for outpatient services.
- Family involvement processes should be clearly delineated and systemized in a protocol that defines their level of involvement, training, duration, and expectations.
- Behavioral interventions are based on the principles of behavior change and aim to decrease difficult behaviors, while teaching alternative behavioral options that the child can employ in challenging situations.
- There is a range of approaches characterized by the different methods they employ to reach identified objectives. These approaches span from traditional behavioral methods to more modern developmental styles that can be characterized as socially pragmatic.
- Various factors and preferences must be considered when choosing the right approach. These include those related to the family's values, resources, and cultural traditions and those related to the child's linguistic and social development, style of learning, range of behaviors, and communicative requirements.

Occupational Therapy Intervention

- Thirty to sixty minutes of face-to-face interaction with the individual or family/caregivers.
- Interventions can focus particularly on engaging the child socially, improving his/her occupational performance and adaptive behaviors, or addressing other appropriate familial priorities.
- Environmental modification to address behavioral, sensory, and functional problems.
- Liaise with the speech and language pathologist (SLP) regarding child intervention programs and arrange co-intervention sessions to promote social interaction.

- Sensory and behavioral strategies targeting improving the child's behavior should be developed and implemented in an integrated way through liaison with their behavioral therapist.

Speech and Language Intervention

- Intervention approaches should focus on building on strengths and reducing challenges related to ASD.
- They should enable the child to develop or modify their skills and communication strategies in ways that enhance their participation in activities.
- Changing environmental factors that present barriers to communication while building on those that improve it includes identifying and using suitable augmentative means/accommodations.
- Similar environmental modifications should be used to encourage appropriate nutritional intake while removing impediments to positive nutritional decisions.

Table 4. Recommendations for Monitoring

Medical Monitoring

- Systems for monitoring progress should be created for each symptom targeted.
- The child's body mass index (BMI) should be measured on standard growth charts and attempts made to pre-emptively modify any identified risks through appropriate advice.
- All prescribed medications should begin with low doses and be subject to monitoring to identify any side effects.
- Use of evidence-based guidelines, for example, those produced by the Canadian Alliance for Monitoring Effectiveness and Safety of Antipsychotics in Children (CAMESA), to monitor metabolic side effects [14].

Clinical Psychology Monitoring

- Regular follow-up assessments of the child's skills and progress.
- Monitor progress of the child's skills and refer for further assessment or services, internal or external, when needed.
- Follow up on emerging child or family concerns or difficulties.
- Plan ahead for major transitions for children and families.
- Monitor and support children and families for emerging comorbid conditions or disorders.

Behavioral Monitoring

- Reliability and validity of the data are consistently measured on the client's performance.
- Evidence of progress report every 6 months for 80–100% of the cases.

Occupational Therapy Monitoring

- Formal re-evaluation is conducted when the OT identifies (in his/her professional opinion) the emergence of new clinical findings, significant changes in the child's condition that necessitate further testing and measures, or insufficient reaction to the care plan. Formal re-evaluation is also conducted when extra information is needed pre-discharge.

Speech and Language Monitoring

- Progress notes are written at intervals that may be stipulated by the facility and report progress on long- and short-term goals.

Table 5. Family Involvement and Family-Centered Practice

Family Involvement

- Engage the family and other stakeholders in a collaborative approach regarding the referral and assessment processes.

- Include the family/caregivers and patient in shared decision making that considers their goals and values.
- Family should be involved in the generalization of the acquired skills.
- The child's needs (and, by extension, family members, etc.) will change over time and differ according to context.
- Parent-mediated interventions should be considered for children and teenagers on the autism spectrum to promote family engagement, family empowerment, and parental satisfaction.

Family-Centered Practice

- The goal of family-centered practice is to develop a partnership so that the family/caregivers fully participate in all aspects of the individual.
- Participation of families/caregivers in services for individuals on the autism spectrum can help to reduce the stress experienced by family members.
- Support may take different forms at different times. It may include coordinating services for the family/caregivers, organizing resources and providing information, teaching the family/caregivers or other significant communication skills and strategies, providing learning opportunities, and advocating for or with the family.